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manchodu@Srini:~/OSLAB$ nano m29.c
manchodu@Srini:~/OSLAB$ gcc m29.c -o m29
manchodu@Srini:~/OSLAB$ ./m29
```

```
=====
                        APPEND REDIRECTION
=====
Enter output filename: m15
Enter message to append: Nice bro

Data appended successfully.
manchodu@Srini:~/OSLAB$
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024

int main()
{
    char filename[MAX_INPUT];
    char message[MAX_INPUT];

    int fd;
    pid_t pid;

    printf("=====\n");
    printf("        APPEND REDIRECTION\n");
    printf("=====\n");

    printf("Enter output filename: ");

    if (fgets(filename,
                sizeof(filename),
                stdin) == NULL)
    {
        return 1;
    }

    filename[strcspn(filename, "\n")] = '\0';

    printf("Enter message to append: ");

    if (fgets(message,
                sizeof(message),
                stdin) == NULL)
    {
        return 1;
    }

    message[strcspn(message, "\n")] = '\0';

    /*
     * O_APPEND ensures that every write
     * occurs at the end of the file.
     */
}
```

```

printf("Enter output filename: ");

if (fgets(filename,
           sizeof(filename),
           stdin) == NULL)
{
    return 1;
}

filename[strcspn(filename, "\n")] = '\0';

printf("Enter message to append: ");

if (fgets(message,
           sizeof(message),
           stdin) == NULL)
{
    return 1;
}

message[strcspn(message, "\n")] = '\0';

/*
 * O_APPEND ensures that every write
 * occurs at the end of the file.
 *
 * Existing data is preserved.
 */
fd = open(
    filename,
    O_WRONLY | O_CREAT | O_APPEND,
    0644
);

if (fd == -1)
{
    perror("open");
    return 1;
}

pid = fork();

if (pid < 0)

```

```

    {
        perror("open");
        return 1;
    }

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        close(fd);
        return 1;
    }

    if (pid == 0)
    {
        /*
         * Redirect stdout to append file.
         */
        if (dup2(fd, STDOUT_FILENO) == -1)
        {
            perror("dup2");
            close(fd);
            exit(EXIT_FAILURE);
        }

        close(fd);

        printf("%s\n", message);

        fflush(stdout);

        exit(EXIT_SUCCESS);
    }

    close(fd);

    waitpid(pid, NULL, 0);

    printf("\nData appended successfully.\n");

    return 0;
}

```

```

manchodu@Srini:~/OSLAB$ nano m30.c
manchodu@Srini:~/OSLAB$ gcc m30.c -o m30
manchodu@Srini:~/OSLAB$ ./m30
=====
                STDERR REDIRECTION
=====
Enter error output filename: odu

Error redirection completed.
Check the specified error file.
manchodu@Srini:~/OSLAB$ |

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```

    fd = open(
        filename,
        O_WRONLY | O_CREAT | O_TRUNC,
        0644
    );

    if (fd == -1)
    {
        perror("open error file");
        return 1;
    }

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        close(fd);
        return 1;
    }

    if (pid == 0)
    {
        /*
         * Redirect standard error (file descriptor 2)
         * to the file.
         */
        if (dup2(fd, STDERR_FILENO) == -1)
        {
            perror("dup2");
            close(fd);
            exit(EXIT_FAILURE);
        }

        close(fd);

        /*
         * Generate an error.
         */
        FILE *file = fopen(
            "file_that_does_not_exist.txt",
            "r"

```

```

/*
 * Redirect standard error (file descriptor 2)
 * to the file.
 */
if (dup2(fd, STDERR_FILENO) == -1)
{
    perror("dup2");
    close(fd);
    exit(EXIT_FAILURE);
}

close(fd);

/*
 * Generate an error.
 */
FILE *file = fopen(
    "file_that_does_not_exist.txt",
    "r"
);

if (file == NULL)
{
    perror("ERROR");
}

/*
 * Generate another error.
 */
int test_fd = open(
    "/invalid/path/test.txt",
    O_RDONLY
);

if (test_fd == -1)
{
    perror("open");
}

exit(EXIT_FAILURE);
}

close(fd);

waitpid(pid, NULL, 0);

```

```

        FILE *file = fopen(
            "file_that_does_not_exist.txt",
            "r"
        );

        if (file == NULL)
        {
            perror("ERROR");
        }

        /*
         * Generate another error.
         */
        int test_fd = open(
            "/invalid/path/test.txt",
            O_RDONLY
        );

        if (test_fd == -1)
        {
            perror("open");
        }

        exit(EXIT_FAILURE);
    }

    close(fd);

    waitpid(pid, NULL, 0);

    printf("\nError redirection completed.\n");
    printf("Check the specified error file.\n");

    return 0;
}

```